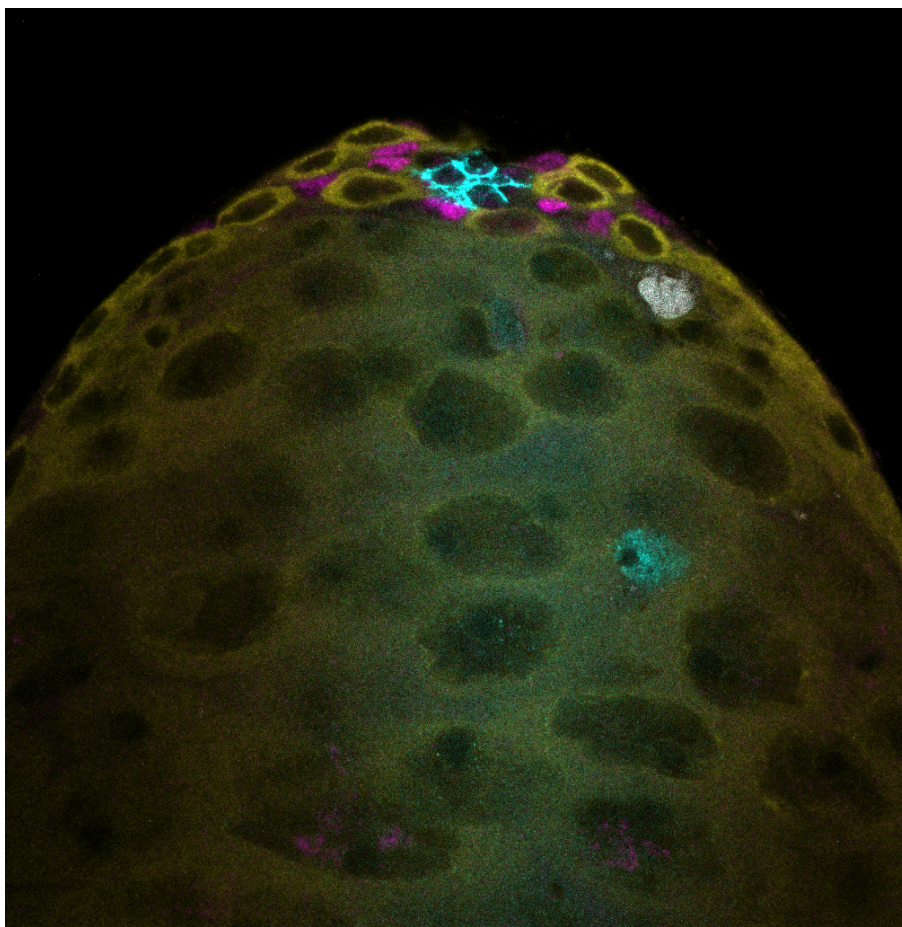


The Role of Monocarboxylate Transporters in Germ Cell Survival in the Drosophila Testis



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Abstract

Cell-cell communication is essential for tissue homeostasis and differentiation. In *Drosophila* testis, the metabolic interplay between somatic cyst cells and germline cells remains largely unexplored. Recent findings suggest that cyst cells provide metabolic support to germ cells, particularly through the transport of lactate via monocarboxylate transporters (MCT). While CG8034 has been identified as the MCT responsible for lactate export from cyst cells, the transporter mediating lactate import into germ cells remains unknown. This study employs the Bam-Gal4/UAS-RNAi system to knock down candidate MCTs in differentiating germ cells and assess their role in germline survival. Immunohistochemistry and lysotracker staining were used to quantify germ cell death across experimental groups. Among the MCTs screened, knockdown of *Hermes* and *Karmoisin* significantly increased germ cell death, indicating their potential roles in lactate transport. Further validation using a second RNAi line and mutant clones is necessary to confirm these findings. Additionally, to elucidate the metabolic fate of lactate in germ cells, preliminary findings in the lab have demonstrated that lactate is converted into malate for downstream metabolism. These findings highlight a critical metabolic coupling between somatic cysts and germ cells, where lactate transport via *Hermes* and *Karmoisin* supports germline survival. This study provides novel insights into the metabolic dependencies of germ cells and contributes to our understanding of inter-lineage metabolic communication.

Declaration of Contribution

I declare that all experiments and the writing of the thesis are my own work, conducted under the supervision and guidance of Dr. Amoyel and Dr. Sainz de la Maza. I was solely responsible for setting up and maintaining my own crosses and stocks, as well as conducting all experiments and data analysis. Dr. Sainz de la Maza and Dr. Amoyel provided instructions and guidance/feedback on the writing of the dissertation.

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1. Introduction

Cell-cell communication is fundamental for the proper function of multicellular organisms, influencing tissue homeostasis, apoptosis, proliferation, and differentiation. Through either direct cell contact or ligand-receptor interaction, cell-cell communication induces intracellular signaling cascades and results in relevant cellular responses, such as changes in gene expression ([Su et al., 2024](#)). Moreover, metabolic communications between cells are also critical for tissue survival and maintenance. The *Drosophila* testis serves as an excellent model for investigating the intricate communication between cells and with their surrounding environment, offering insights into how various signals and metabolic cues are coordinated to determine cell fate and influence cell behavior.

1.1. *Drosophila* Testis as a Model System

1.1.1. Overview of *Drosophila Melanogaster*

Drosophila melanogaster, commonly known as the fruit fly, has been used as a versatile model organism in various areas of biological research, including genetics, development, cancer, neuroscience, aging, etc. With a genome of ~180 Mb and 13000 protein-coding genes organized into four chromosomes, *Drosophila* offers a small, compact, and well-mapped genome. Its genome can be easily manipulated by using genetic tools such as RNAi, CRISPR/Cas9, the Gal4/UAS system, etc. Additionally, *Drosophila* shares many common biological processes and orthologous genes with humans, allowing insights into human diseases through the study of these orthologue genes and pathways. What's more, *Drosophila* undergoes a rapid lifecycle containing four distinct stages: egg, larva, pupa, and adult, which takes only 10 days to complete at 25 °C ([Jennings, 2011](#)). As a result, the small genome, genetic manipulability, conserved biological pathways, and short lifecycle make *Drosophila* an ideal model organism for large-scale genetic screens and experiments.

1.1.2. Stem Cell Niche and Communications in the Drosophila Testis

Stem cells are a unique group of cells characterized by their ability to self-renew and differentiate, which are critical in maintaining tissue homeostasis, replacement, and growth. After each division, daughter cells can either retain stem cell identity or differentiate into a functional-specific lineage. This balance is tightly regulated by signal inputs from the surrounding microenvironment, for example, the stem cell niche provides signals for stem cells to retain their self-renewal capacity ([Jones and Wagers, 2008](#)). In the Drosophila testes, the stem cell niche is located at the apical tip ([Figure 1A](#)) of the testis, containing 10-15 non-dividing stromal cells that form the hub. Surrounding the hub, two types of stem cells are present: germline stem cells (GSCs) and cyst stem cells (CySCs). GSCs divide asymmetrically and produce two daughter cells: one daughter cell remains in contact with the hub, retaining the stem cell identity; the other one differentiates into a gonialblast, the sperm progenitor. Each gonialblast, along with its progeny, is flanked and encapsulated by two postmitotic cyst cells (CCs) derived from CySCs ([de Cuevas and Matunis, 2011](#)). Before entering meiosis and terminal differentiation, each gonialblast undergoes four rounds of synchronous transit-amplifying (TA) division, producing 16 interconnected spermatogonial cells ([Lim and Fuller, 2012](#)) ([Figure 1G](#)). These different types of cells can be easily identified and visualized using immunohistochemistry ([Figure 1B-F](#)).

The hub cells act as the source of local signals to provide cues to maintain the stem cell fate of both stem cell lineages. Janus kinase and Signal Transducer and Activation of Transcription (JAK/STAT) signaling are activated in response to the Unpaired (Upd) ligand from the hub, which promotes self-renewal predominantly in the CySC lineage by activating the expression of transcription factor Zinc-finger homeodomain 1 (Zfh1) ([Leatherman and DiNardo, 2008](#)). Other than JAK/STAT, CySC maintenance also requires Hedgehog (Hh), Hippo (Hpo), Slit-Robo, and Mitogen-Activated Protein Kinase

(MAPK) signaling in parallel ([Michel et al., 2012](#), [Amoyel et al., 2014](#), [Amoyel et al., 2016a](#), [Stine et al., 2014](#), [Amoyel et al., 2013](#)). As CySC progenies differentiate and move away from the hub, *Zfh1* is downregulated, while Eyes Absent (*Eya*) is upregulated. The differentiation is stimulated by the PI3K/Tor pathway, which is upregulated in CySC daughter cells not in contact with the hub ([Amoyel et al., 2016b](#)). However, Tor signaling only primes the CySC to a permissive differentiation state, and CySCs do not commit to differentiation until associated with a gonialblast ([Yuen et al., 2021](#)). For GSCs, JAK/STAT signaling mainly contributes to maintaining the attachment of GSC to the hub ([Greenspan et al., 2015](#), [Leatherman and DiNardo, 2010](#)). The maintenance of the GSC identity is governed by BMP signaling from the niche and the surrounding cyst cells ([Figure 1H](#)). Studies using temperature-sensitive *glass bottom boat* (*gbb*) and *decapentaplegic* (*dpp*) mutants (BMP2/4 orthologs in *Drosophila*) have shown that the loss of BMP signaling results in GSC depletion, as the BMP signals are important in keeping the pro-differentiation factor *Bam* repressed and preventing premature GSC differentiation ([Kawase et al., 2004](#), [Shivdasani and Ingham, 2003](#)). Furthermore, Epidermal Growth Factor Receptor (EGFR) signaling in cyst cells is required to restrict self-renewal of GSCs and promote differentiation. Loss of EGFR signaling resulted in the accumulation of undifferentiated early germ cells throughout the testis, along with clusters of 16 or more spermatogonia resulting from failure to cease mitotic division. These findings revealed the role of EGFR signaling in both GSC-gonialblast and spermatogonia-spermatocyte differentiation ([Kiger et al., 2000](#)). Therefore, both signals from the hub and the intercommunication between the two stem cell lineages are critical for cell fate determination and development.

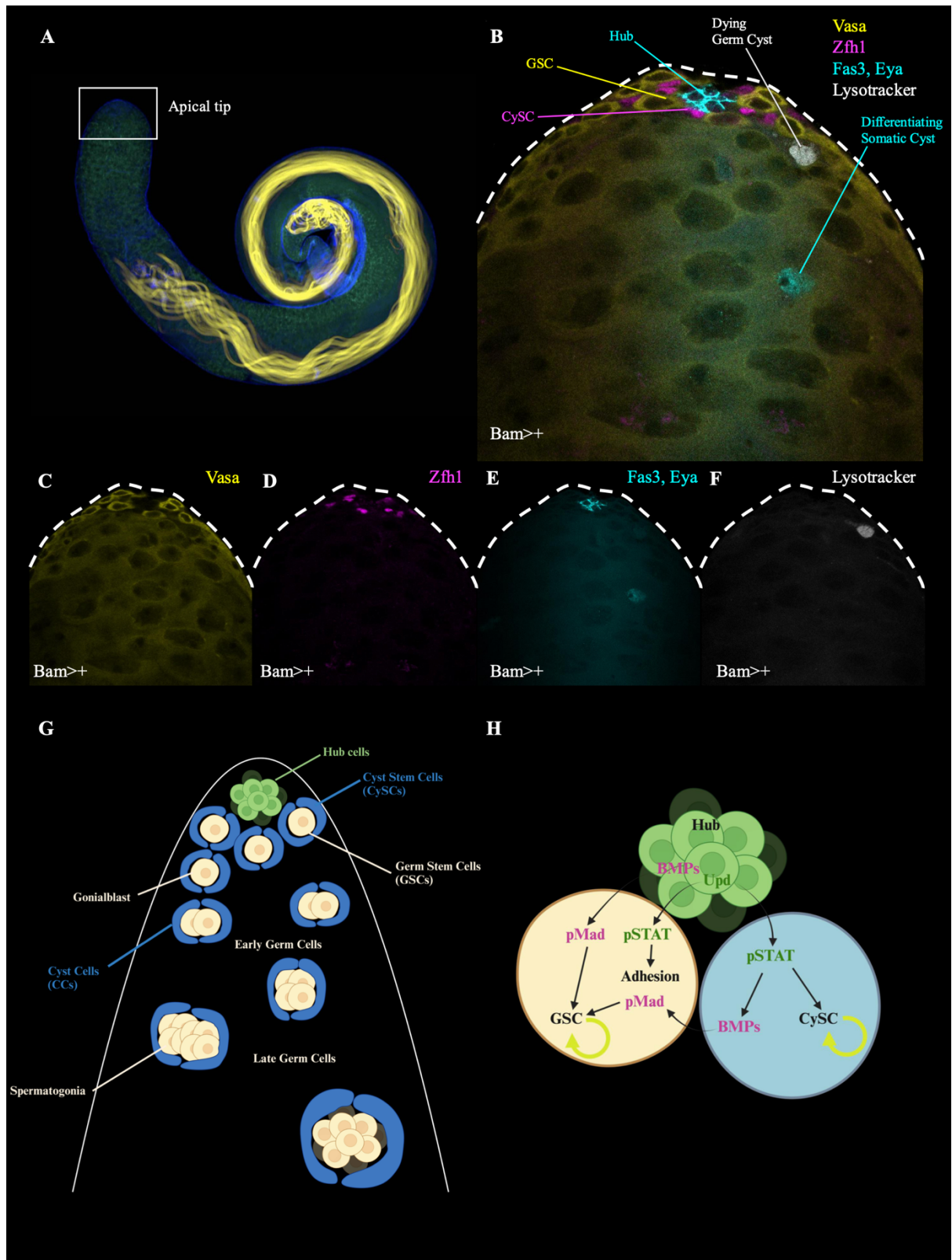


Figure 1 Schematic of the *Drosophila* testis and the Intercellular Signaling Pathway. (A) *Drosophila* spiral-shaped testis, 150x ([Laurinyecz, 2013](#)). (B-F) Immunostaining of the testis: anti-Vasa (yellow) for germline cells; anti-Zfh1 (magenta) for cyst stem cells; anti-Fas3 and anti-Eya (cyan) for hub cells and late differentiating somatic cyst cells, respectively; lysotracker dye (white) for dying germ cells. (G) Schematics of the *Drosophila* Testis. (H) Signaling communication in *Drosophila* Testis (adapted from Dr. Amoyel's slides).

1.2. Metabolic Regulation and Cross-Lineage Support in *Drosophila* Testis

Metabolism is a set of biochemical processes that sustains life by breaking nutrients into small molecules, releasing energy (catabolism), and synthesizing building blocks of proteins, lipids, nucleic acids, and carbohydrates using energy (anabolism). In the *Drosophila* testis, the metabolic profile of somatic cysts changes dynamically during differentiation. It is hypothesized that these metabolic shifts not only adapt to the changing cellular demands of somatic cells themselves but also support the metabolic requirements of the differentiating germ cysts.

Glycolysis is a cytoplasmic process that breaks down glucose into pyruvate, yielding energy in the form of ATP and NADH, while oxidative phosphorylation occurs in mitochondria, using oxygen to generate ATP through the electron transport chain (ETC) and chemiosmosis. In *Drosophila*, trehalose, a non-reducing disaccharide, serves as the major sugar source in the hemolymph, which is the circulating fluid of insects and is equivalent to vertebrate blood. Trehalose is transported into the CCs via Trehalose transporters (Tret) and hydrolyzed to glucose by Trehalase (Treh) before entering glycolysis ([Figure 2A](#)). As the cyst cell differentiates, the metabolism profile of cyst cells changes as the metabolic gene expression changes. To investigate this, RNA sequencing was performed, followed by transcriptome and gene ontology (GO) analysis. After the identification of the differentially expressed genes in differentiated CCs and CySCs, these genes were mapped to GO terms based on their associated biological pathways. GO analysis revealed that the differentiated CCs upregulate multiple core enzymes involved in glycolysis and oxidative phosphorylation, including the TCA cycle, pyruvate metabolism, and the ETC, as well as proteins involved in mitochondria biogenesis and function. Besides, measurements of increased mitochondria activity using the MitoTimer sensor demonstrate a metabolic shift towards oxidative phosphorylation in CCs ([Hof-Michel and Bökel.,](#)

[2020, de la Maza et al., 2022](#)). This was also validated in vivo using reporters (Sainz de la Maza, unpublished results), which demonstrated that the levels of key glycolytic enzymes, including Aldolase 1 (Ald1), Pyruvate Kinase (PyK), and Lactate Dehydrogenase (Ldh), are elevated in differentiated somatic cells compared to CySCs. By knocking down glycolytic enzymes in the CCs using Tj/Gal4, it was revealed that glycolysis is not required for the survival and differentiation of the CCs, and CCs rely more on oxidative phosphorylation. Moreover, glycolysis is also not required in early germ cells and differentiating germ cells by knocking down relative enzymes using Nanos/Gal4 and Bam/Gal4. However, when knocking down the glycolytic enzymes in CCs, an increased number of dying germline cysts was observed. Accordingly, the hypothesis that increased glycolysis in CCs is used to metabolically support the germline is raised ([Figure 2B](#)).

Metabolites produced by glycolysis can either be shuttled to support germ cysts or transported to mitochondria in the somatic CCs for oxidative phosphorylation ([Figure 2B](#)). To further validate the metabolite's role in germline survival and investigate which intermediate metabolite was shuttled, previous work in the lab used RNAi to silence genes associated with somatic metabolism and evaluate germline survival. Intriguingly, silencing the gene encoding for mitochondria pyruvate carrier Mpc1/2 in the somatic cells leads to a decrease in the number of dying germ cells, indicating the retention of pyruvate in CC's cytoplasm is beneficial for germline survival. Similarly, knocking down somatic Pyruvate dehydrogenase (Pdh), the enzyme that converts pyruvate into acetyl-CoA, also results in reduced germline death. On the other hand, downregulation of Pyruvate dehydrogenase kinase (Pdk), the negative regulator of Pdh, causes an increase in germ cell death. Additionally, Ldh knockdown in CCs results in an increased level of pyruvate in the CCs and a decreased level of lactate in both the CCs and the germline cells, accompanied by increased germline death. These findings highlight the critical role of lactate availability in

CCs to provide metabolism support to the germline compartmentally, and lactate is the major metabolite transported to the germ cysts. However, there are still gaps in the study. For example, the transporters that shuttle lactate from CCs into the intercellular cavity and then to the germ cells have yet to be identified.

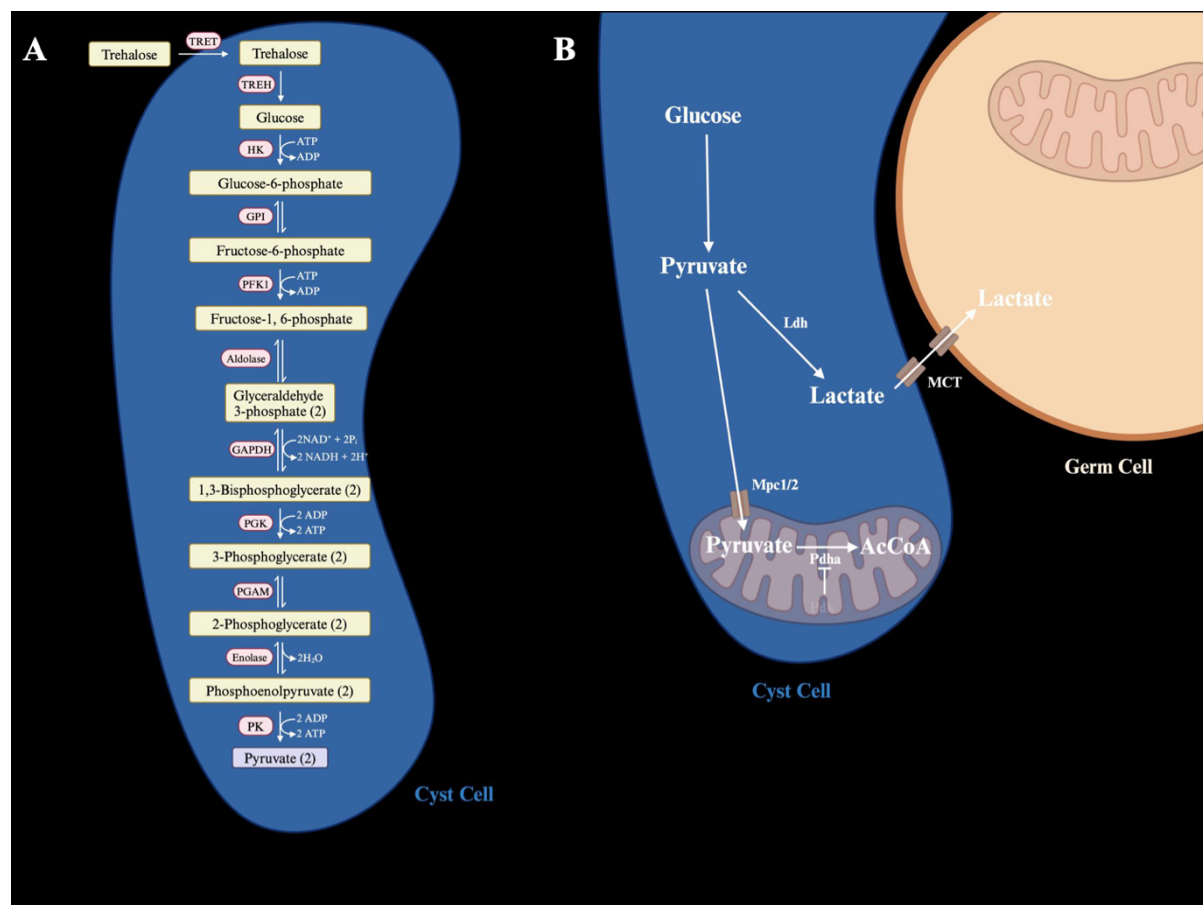


Figure 2 Glycolysis in *Drosophila testis*. (A) Glycolytic pathway in the somatic cysts. (B) Schematic of the metabolic support from somatic cysts to germ cysts (adapted from Dr. Sainz de la Maza's slides).

1.3. Monocarboxylate Transporters in Intercellular Metabolite Exchange

Monocarboxylate transporters (MCTs) are transmembrane carrier proteins that belong to the solute carrier 16 (SLC16) superfamily. These transporters are primarily responsible for the transport of short-chain monocarboxylates, which are small molecules containing a single carboxyl group. These include metabolites essential in energy metabolism, such as pyruvate, lactate, and ketone bodies. The transport process is often coupled with proton symport, making MCTs essential not only for metabolism communication between cells

but also for maintaining pH homeostasis ([Felmlee et al., 2020](#), [Halestrap and Price, 1999](#)). In mammalian spermatogenesis, Sertoli cells exhibit a cancer-cell-like metabolism profile characterized by a heavy reliance on glycolysis. Sertoli cells export pyruvate and lactate into the tubular fluid via MCTs, enabling the germ cells to import these metabolites via their MCTs ([Crisóstomo et al., 2018](#)). Additionally, the Schimeier lab, which works on the *Drosophila* brain, has demonstrated that glial glycolysis is important for neuronal survival. Glial cells, the support cells of neurons, take up trehalose across the blood-brain barrier and fuel the neurons with glycolytic products. Knocking down MCTs in both glial cells and neurons results in neurodegeneration ([Volkenhoff et al., 2015](#)). Given the critical metabolic support role of the Sertoli cells and glial cells to germ cells and neurons, respectively, and their functional analogies with CCs, we hypothesize that the metabolism communication between CCs and the germ cells in *Drosophila* testis is critical and is similarly mediated by MCTs.

Based on the MCT list from FlyBase and Volkenhoff et al., previous lab work used *tj-Gal4/UAS-RNAi* to knock down each MCT on the list in somatic cells, aiming to identify the MCT involved in the process of transporting metabolites from the somatic cell into the intercellular cavity. Fly Cell Atlas data guided the focus toward five MCTs that show expression in the CCs. Among these, preliminary results have identified CG8034 as the most probable candidate, as its knockdown leads to a statistically significant increase in germ cell deaths compared to control. Further investigation to validate CG8034 using a second RNAi or mutant clone is still ongoing.

1.4. Research Aim

The lab has been focused on investigating the metabolic interplay between somatic cysts and germ cells. Recently, it has been discovered that the glycolytic product lactate is

transported from the somatic cysts to the germ cells to support their survival. The next step aims to identify the transporter that shuttles the metabolite and its role in metabolic support. Building on prior work that screened the MCTs in the somatic cysts, this project focuses on identifying the MCTs in the germline that shuttle metabolites from the intercellular cavity into the differentiating germ cells using genetic screens.

2. Material and Methods

2.1. Materials

2.1.1. Fly Stocks

Genotype	Targeted Gene	Source	Identifier
<i>D. melanogaster</i> : <i>Sa-GFP</i> ; <i>Bam-Gal4</i> , <i>UAS-Dicer2</i> (<i>TM3, sb</i>)	N/A	Amoyel Lab	N/A
<i>D. melanogaster</i> : <i>yw122</i> ; +; +	N/A	Amoyel Lab	N/A
<i>D. melanogaster</i> : <i>w1118</i> ; <i>P{GD877}</i> v7292	<i>CG10019</i>	Vienna Drosophila Resource Center (VDRC)	VDRC: 7292; Flybase: FBst0470632
<i>D. melanogaster</i> : <i>w1118</i> ; <i>P{GD1807}</i> v7314	<i>Hermes</i>	VDRC	VDRC: 7314; Flybase: FBst0470641
<i>D. melanogaster</i> : <i>P{KK101781}</i> VIE-260B	<i>Karmoisin</i>	VDRC	VDRC: 105429; Flybase: FBst0477256
<i>D. melanogaster</i> : <i>P{KK105659}</i> VIE-260B	<i>Tarag</i>	VDRC	VDRC: 101290; Flybase: FBst0473163
<i>D. melanogaster</i> : <i>w1118</i> ; <i>P{GD1995}</i> v39545/ <i>TM3</i>	<i>Tarag</i>	VDRC	VDRC: 39545; Flybase: FBst0463082
<i>D. melanogaster</i> : <i>w1118</i> ; <i>P{GD3405}</i> v45188/ <i>TM3</i>	<i>CG16700</i>	VDRC	VDRC: 45188; Flybase: FBst0465974
<i>D. melanogaster</i> : <i>P{KK105945}</i> VIE-260B	<i>CG16700</i>	VDRC	VDRC: 110058; Flybase: FBst0481652
<i>D. melanogaster</i> : <i>w1118</i> ; <i>P{GD3448}</i> v51157	<i>Outsiders</i>	VDRC	VDRC: 51157; Flybase: FBst0469314
<i>D. melanogaster</i> : <i>w1118</i> ; <i>P{GD2022}</i> v8976/ <i>CyO</i>	<i>CG8389</i>	VDRC	VDRC: 8976; Flybase: FBst0471318
<i>D. melanogaster</i> : <i>P{KK106998}</i> VIE-260B	<i>CG8925</i>	VDRC	VDRC: 101128 Flybase: FBst0473001

<i>D. melanogaster</i> : P{KK110059}VIE-260B	CG13907	VDRC	VDRC: 107339; Flybase: FBst0479161
<i>D. melanogaster</i> : P{KK108618}VIE-260B	CG3456	VDRC	VDRC: 106773; Flybase: FBst0478597
<i>D. melanogaster</i> : P{KK111485}VIE-260B	CG6006	VDRC	VDRC: 106513; Flybase: FBst0478337
<i>D. melanogaster</i> : w1118; P{GD1940}v4607	Silnoon	VDRC	VDRC: 4607; Flybase: FBst0466501
<i>D. melanogaster</i> : w1118; P{GD1940}v4609	Silnoon	VDRC	VDRC: 4609; Flybase: FBst0466509
<i>D. melanogaster</i> : w1118; P{GD1648}v6452/TM3	CG8468	VDRC	VDRC: 6452; Flybase: FBst0470322

Table 1 Fly stocks used for this project

2.1.2. Antibodies used

Antibodies	Source	Catalog Number
Mouse monoclonal anti-Eya (1:20)	Developmental Studies Hybridoma Bank (DSHB)	Cat# eya10h6; RRID: AB_528232
Mouse monoclonal anti-Fas3 (1:20)	DSHB	Cat# 7G10; RRID: AB_528238
Rat monoclonal anti-Vasa (1:20)	DSHB	Cat# anti-vasa; RRID: AB_760351
Rabbit anti-zfh1 (1:500)	Amoyel Lab	N/A
Donkey anti-Rat IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor™ 488 (1:200)	Thermo Fisher Scientific	Cat# A-21208; RRID: AB_2535794
Alexa Fluor® 647-conjugated AffiniPure® Donkey Anti-Rabbit IgG (H+L) (1:200)	Strattech	Cat# 711-605-152; RRID: AB_2492288
Donkey anti-Mouse IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor™ Plus 405 (1:200)	Thermo Fisher Scientific	Cat# A48257; RRID AB_2884884

Table 2 Primary and Secondary antibodies used for this project

2.1.3. Chemical Reagents

Chemical Reagents	Source	Catalog Number
Schneider's Insect Medium	Sigma	S0146
Pierce 16% Formaldehyde (w/v), Methanol-free (PFA)	Fisher Scientific	11586711
Phosphate Buffered Saline 10X (PBS)	Fisher Scientific	10649743
Triton X-100	Fisher Scientific	10102913
Bovine Serum Albumin (BSA)	Sigma	A9647
LysoTracker™ Red DND-99, special packaging	ThermoFisher	L7528
SlowFade Gold antifade reagent	Fisher Scientific	S36936

Table 3 Chemical Reagents used for this project

2.1.4. Software and Algorithms

Software and algorithms	Source	Link
Image J	(Schneider et al., 2012)	https://imagej.net/ij/
Prism	Graph Pad	https://www.graphpad.com/
Excel	Microsoft	https://www.microsoft.com/en-gb/microsoft-365/excel

Table 4 Softwares used in this project (Schneider et al., 2012)

2.2. Methods

2.2.1. Genetic Tools

The Gal4/UAS system is a powerful genetic tool used to control gene expression in a spatially and temporally regulated manner ([Figure 3A](#)). The Gal4/UAS system is bipartite, consisting of two key components: Gal4, a transcriptional factor derived from yeast; UAS (Upstream Activation Sequence), a DNA sequence that Gal4 binds to, activating the expression of the gene of interest ([Caygill and Brand, 2016](#)). In this system, a transgenic Gal4 driver line, where Gal4 expression is controlled by a specific promoter, is crossed with a transgenic UAS line containing the gene of interest. The resulting progeny expresses the gene in cells where the Gal4 driver is active. By selecting different promoters to control Gal4 expression, researchers can target specific cell types or stages of development, enabling precise genetic manipulation. In *Drosophila* testis, for example, *Bam* (*bag of*

marbles) is a germ cell differentiation marker that is turned on at the 4-cell stage ([McKearin and Ohlstein, 1995](#)), hence Bam-Gal4/UAS drives the expression of the gene of interest in differentiating germ cells. Other prominent drivers include Upd-Gal4, Nanos-Gal4, and Traffic jam-Gal4, which drive gene expression in hub cells, GSCs, and somatic CCs, respectively ([Rodriguez et al., 2012](#), [Brand and Perrimon, 1993](#)).

What's more, Gal80, a negative regulator of the Gal4/UAS system, binds to the activation domain of Gal4, inhibiting its transcription-promoting activity ([Pfeiffer et al., 2010](#)). Additionally, Gal80 has a temperature-sensitive variant (Gal80^{Ts}) that is active at the permissive temperature, blocking Gal4 expression and successive gene expression. At restrictive temperatures, Gal80^{Ts} become inactive, allowing Gal4 to induce gene expression ([Zeidler et al., 2004](#)) ([Figure 3B](#)). This enables temporal control of gene expression for conditional knockdowns.

The balancer chromosome is another important genetic tool in *Drosophila* ([Figure 3C](#)). Balancer chromosomes always contain multiple inversions inhibiting homologous recombination during meiosis, thus preventing genetic mix. Balancers usually have a dominant phenotypic marker that helps researchers easily distinguish flies carrying balancers from those without it. Moreover, most balancers are homozygous lethal, which ensures that mutant alleles or transgene insertions are maintained either heterozygously or homozygously in a population.

Since the discovery of the GAL4 system, thousands of Gal4 driver lines have been developed, along with temperature-sensitive Gal80 and RNAi techniques, enabling tissue- and time-specific gene manipulation.

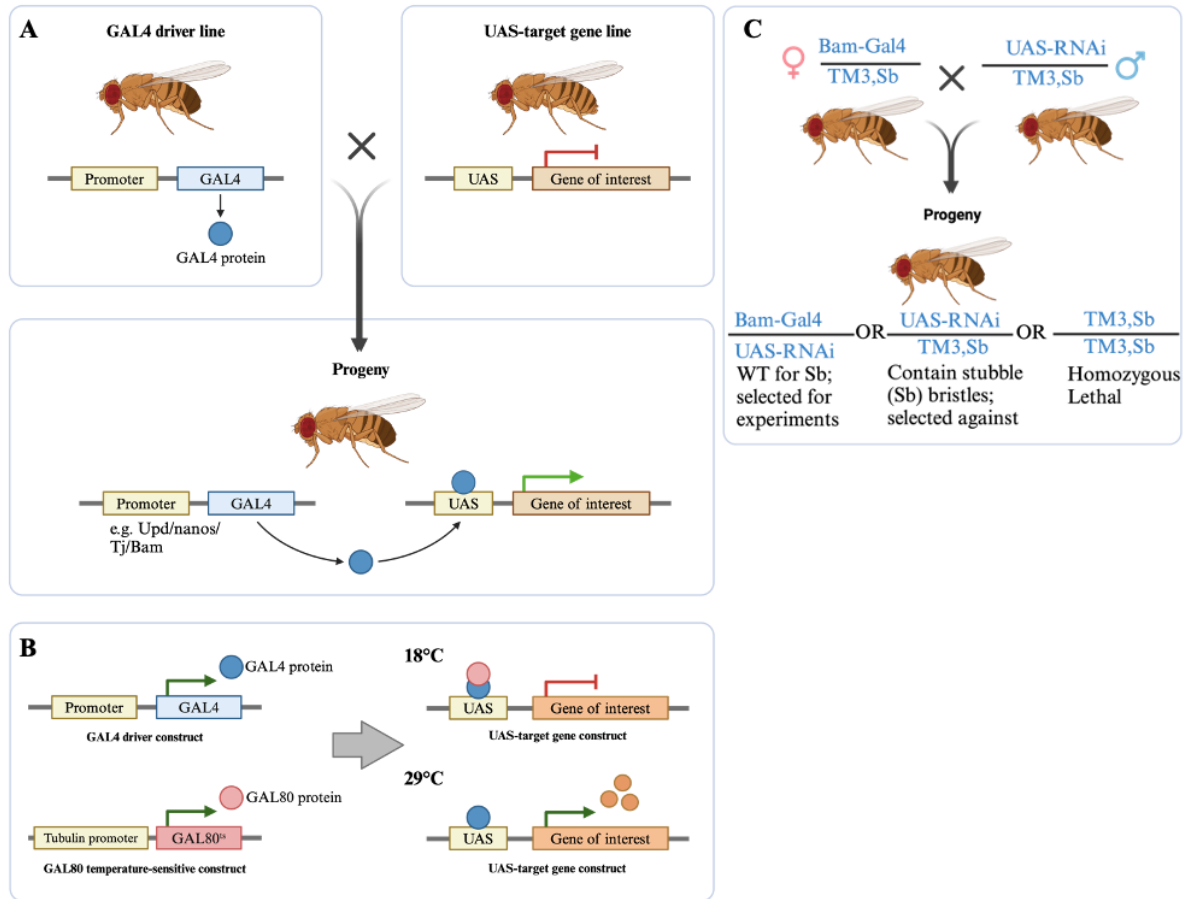


Figure 3 Genetic Tools in *Drosophila*. (A) The Gal4-UAS system. (B) Temperature-sensitive Gal4/Gal80-UAS system. (C) Balancer chromosome segregation and genetic crosses

2.2.2. Fly Husbandry

To conduct lineage-wide MCT knockdown experiments, Bam-Gal4 virgin females (*sa-GFP*; *Bam-Gal4*, *UAS-Dicer2*/(*TM3*)) were identified and picked within 0-10 hours after eclosion and mated with UAS-RNAi males (Table 1). To optimize fertility, young UAS-RNAi males were selected, and a male-to-female ratio of 1:3 was maintained across crosses. The control was set up the same way using the Bam-Gal4 virgin females with wild-type males (*yw*122; +; +). To maintain the health of the flies, both the crosses and control were flipped into fresh vials every 4 days. All crosses and Bam-Gal4 stocks were kept at room temperature (RT), and the UAS-RNAi stocks were kept at 18 °C to slow down their life cycle. After around 14 days in RT, the F1 progeny males were selected against the balancers and collected within 0-3 days after eclosion. Then, these males were transferred to 29 °C

for 10 days before dissection to enhance Gal4 activity, thereby ensuring stronger RNAi-mediated gene silencing.

2.2.3. Immunohistochemistry

After 10 days at 29 °C, males were dissected in Schneider's medium to maintain tissue viability. The abdomens containing the testes were transferred to Eppendorf tubes with 750 µl Schneider medium and LysoTracker Red (1:1000, 0.75 µl) with shaking for 1 hour. Samples were then fixed in 4% formaldehyde in PBS (diluted from 16% PFA) for 15 minutes at RT with shaking, which preserves the tissues by stabilizing their cellular structure and preventing degradation. Following fixation, the samples were washed twice with 0.5% PBS-Triton (PBS-T) for 30 minutes to remove residual PFA and permeabilize cell membranes for antibody entry. Blocking was performed using PBTB (1X PBS, 0.2% Triton, 1% BSA) for 1h at RT to remove nonspecific binding. Then, the samples were incubated overnight in primary antibodies ([Table 2](#)) diluted in PBTB at 4°C. The next day, the samples were washed twice for 30 minutes in PBTB before incubation with secondary antibodies ([Table 2](#)) diluted in PBTB for 2 hours at RT with shaking. After two final 30-minute washes in 0.2% PBS-T, the testes were mounted in SlowFade antifade reagent on glass slides and covered with coverslips before being imaged using a confocal microscope. All reagents used in this protocol are listed in [Table 3](#).

2.2.4. Confocal Microscopy

Images were acquired using a Zeiss LSM880 Confocal Microscope equipped with a 63x/1.4 Oil DIC M27 objective lens and controlled by ZEN (black edition) software. The excitation wavelengths used to excite the fluorophores were 488 nm (green) for anti-Vasa, 568 nm (red) for LysoTracker Red, 647 nm (far red) for anti-zfh1, and 405 nm (UV) for anti-Eya

and anti-Fas3. For Z-stack settings, the upper and lower limits were manually set for each testis with a Z-stack interval of 2 μm to ensure resolution and depth capture.

2.2.5. Quantification and Statistical Analysis

The acquired images were processed using ImageJ (Fiji), where brightness and contrast were adjusted for each channel to optimize figure quality. Following the adjustment, germ cell death was evaluated by measuring the number of dying germ cells using LysoTracker dye, which is a highly soluble small molecule that is retained in acidic compartments ([Fogel et al., 2012](#)). The LysoTracker-positive germ cells were manually counted. To prevent errors, care was taken to avoid double-counting cells across different confocal planes and to distinguish other small acidic compartments such as lysosomes from whole cells.

After quantification, statistical analyses were performed using Prism 10. Nonparametric one-way ANOVA and t-test were used to compare experimental groups to the control to evaluate whether our experimental groups were significantly different from the control. Nonparametric tests were chosen due to the discrete nature of our data rather than continuous distribution.

3. Results

3.1. Knockdown specific MCTs using Bam-Gal4/UAS-RNAi system

Bam-Gal4/UAS-RNAi is used to knock down genes encoding MCTs in differentiating germ cysts, and the number of germ cell deaths in each experimental group was compared to the control. The decision to use Bam-Gal4, a relatively late germ cell driver, rather than an early germ cell driver like nanos-Gal4 is based on the timing of cyst-germ metabolic communication. Around the 4-cell stage, germ cells are sealed off from the external environment by cyst cells, which form tight junctions around them. This encapsulation prevents germ cells from independently acquiring metabolites. Therefore, cyst-germ metabolic communication is more likely to happen at this stage, coinciding with Bam-Gal4 activity. If the specific MCT is involved in the process of shuttling metabolites from the intercellular cavity into the germline, the germ cell survival will be affected after the knockdown of that particular MCT. The number of dying germ cells indicates the degree of participation of relative MCTs in the process of metabolite shuttling. The selection of specific MCTs for this screen was carefully determined. Firstly, we examined the expression level of individual MCTs in germ cells based on Fly Cell Atlas, leading to nine potential MCTs ([Figure 4](#)). Among these, CG8051 is not included in the study as there is no available RNAi. Moreover, FlyBase predicts the function of each MCT and its potential substrates using single-cell RNA sequencing data, followed by GO analysis and structural & domain characterization. These predictions further refined our selection. For example, Prestin, a member of the SLC26 family of anion exchangers, mediates the transport of chloride, bicarbonate, and sulfate ([Hirata et al., 2012](#)). Although Prestin is expressed in the germ cells, it does not transport metabolites and was therefore excluded from the screen. Conversely, the seven selected MCTs (Hermes, Karmoisin, Tarag, Outsiders, CG8389,

CG10019, and CG8034) are predicted to transport key metabolites, including acetate and pyruvate ([Appendix 1](#)), and are hence prioritized in the screen.

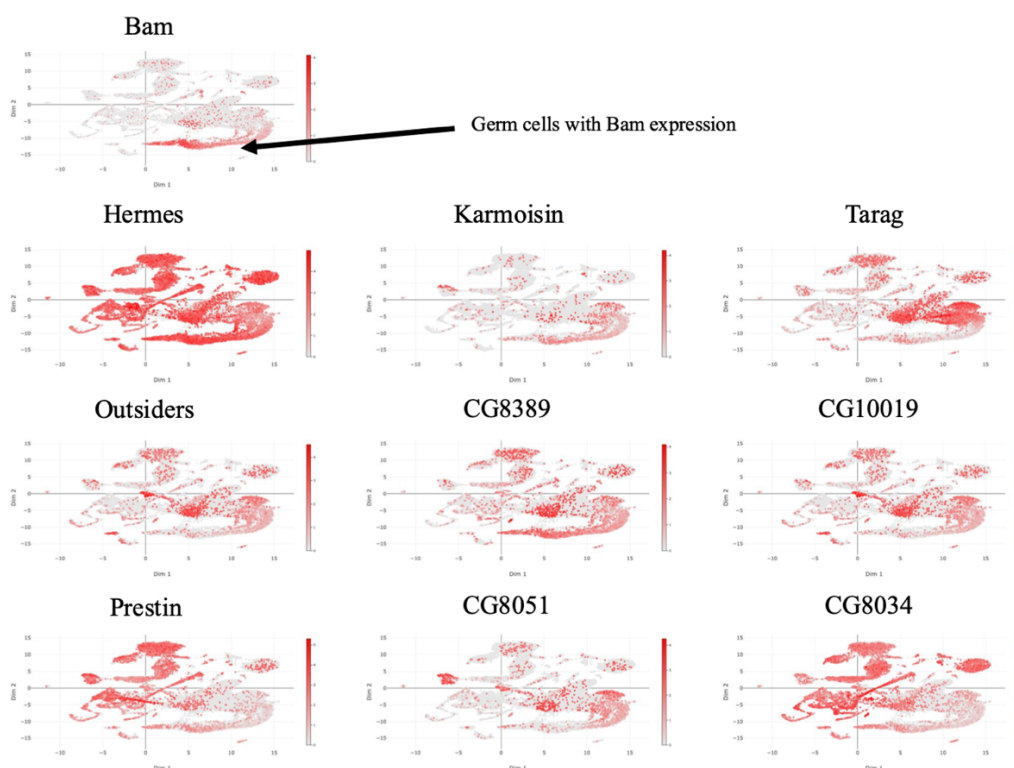


Figure 4 Metabolite transporters expressed in germ cells: UMAP plot of expression levels of genes encoding MCTs in all types of cells in *Drosophila* Testis (From Fly Cell Atlas). The higher dimensional single-cell RNA sequencing data is scaled down to two dimensions, in which similar cells are close to each other, forming clusters. The intensity of red dots represents the expression level, with higher intensity indicating stronger expression in the relevant cluster. The MCT expressions were compared relative to the Bam expression, which serves as a marker for the germ cell population. 9 MCTs that were expressed in the germ cells were listed.

3.2. MCT knockdown in germline does not affect normal *Drosophila* testis development

The targeted knockdown of MCT in the germ cysts was carried out, and representative images of each group are displayed in [Figure 5](#). The hub cells were stained with anti-Fas3 in cyan, indicating the position of the stem cell niche at the apical tip of the testis. The *Zfh-1* positive (magenta), *Eya*-negative cells in direct contact with the niche were identified as CySCs, while those not in direct contact were considered as their immediate daughter cells ([Leatherman and DiNardo, 2008](#)). As the CySCs differentiate, they migrate away from the hub, rapidly losing *zfh-1* expression (magenta) and beginning to express *Eya* (cyan).

The Eya-positive differentiating somatic cysts were observed distal to the niche. The anti-Vasa stains for the germ cells and the Vasa-positive cells in contact with the niche are identified as the GSCs. In the control ([Figure 5A](#)), small Vasa-positive cells are observed around the niche, and as they progressed further down the testis, the Vasa-positive germ cysts enlarged progressively, indicating normal division and differentiation of spermatogonia. All cell types were present in all experimental groups, and similar cell patterns were observed ([Figure 5B-F](#)). There were no significant morphological or numerical alterations in any of the cell types. However, differences were observed when counting the lysotracker-positive cysts in the experimental groups. The number of lysotracker-positive cells varies across different experimental groups, with some showing statistical differences when compared with the control group. In conclusion, MCT knockdowns did not disrupt the normal morphology of different types of cells in the testis and did not impede normal testicular development in *Drosophila*, while some MCT knockdowns influence the germ cell survival, which is reflected by the number of dying differentiating cysts.

3.3. Lysotracker-positive germ cysts in the control and experimental group

In the control group, a basal level of lysotracker-positive germ cysts (white) was observed in the Vasa-positive cyst population further down the testis after the 4-cell stage, where they were no longer in contact with the hub. An average of 6.57 lysotracker-positive cysts per testis (N = 232 testes) was obtained, with a standard deviation (SD) of 2.3. For the experimental groups, lysotracker-positive cysts were similarly observed distal to the hub, exhibiting the same morphological characteristics as the control.

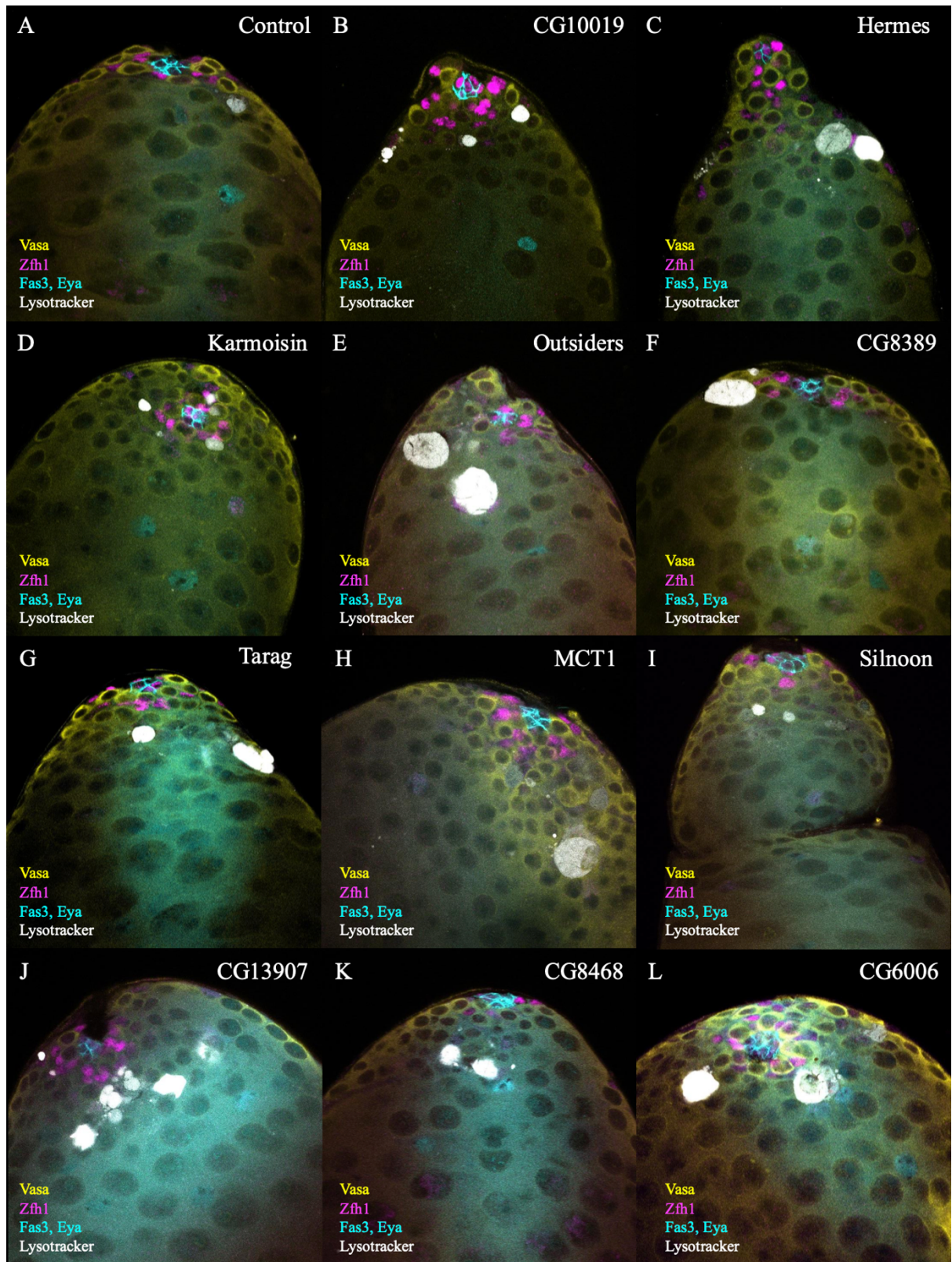


Figure 5 Representative testes of the control and the RNAi knockdown groups. (A) Control. (B) CG10019 RNAi. (C) Hermes RNAi. (D) Karmoisin RNAi. (E) Outsiders RNAi. (F) CG8389 RNAi. (G) Tarag RNAi. (H) MCT1 RNAi. (I) Silnoon RNAi. (J) CG13907 RNAi. (K) CG8468 RNAi. (L) CG6006 RNAi.

3.4. *Hermes* & *Karmoisin* knockdown significantly increase germ cyst death

As the experiments were conducted over a long period of time, it is crucial to first rule out any significant difference in the number of lysotracker-positive cells between control groups collected on different dates. This ensured that the comparisons between controls and experimental groups were not biased by control batch variations. To assess this, the Kruskal-Wallis test, a non-parametric one-way ANOVA test that ranks data points and compares median distribution across multiple groups ($N > 2$), was performed on 12 batches of control data. The test yielded a p-value of 0.5036, failing to reject the null hypothesis, indicating there were no significant differences in the number of lysotracker-positive cysts across control batches ([Figure 6E](#)). This supports a robust comparison of the experimental group with the control group, even when data were collected on different dates.

Following that, lysotracker-positive germ cysts were counted in a minimum of 50 testes ($N > 50$) for each experimental group. Among the seven selected MCTs, CG8034 knockdown was not performed due to time constraints. Additionally, 5 MCTs not expressed in the germ cells (MCT1, CG13907, CG8468, CG6006, and Silnoon) were included in the screen as negative controls. The means and SDs of the number of lysotracker-positive cysts for each group were calculated and shown in [Figure 6A](#). Among the 11 experimental groups, the mean numbers of lysotracker-positive cysts in the *Hermes* and *Karmoisin* knockdown groups were mostly increased with a mean of 8.68 (SD = 2.8) and 7.62 (SD = 2.4), respectively. To further evaluate whether the MCT knockdown-induced increase of germ cell death was statistically significant, a Kruskal-Wallis test was conducted to compare the number of lysotracker-positive germ cysts between experimental groups and the control. As illustrated in [Figure 6B&C](#), *Hermes* and *Karmoisin* knockdown resulted in a significant increase in the number of lysotracker-positive cysts, with p-values of <0.0001 for *Hermes*

and 0.0313 for *Karmoisin*. Since both p-values are below the significance threshold of 0.05, the null hypothesis is rejected, confirming a statistically significant increase.

For all the other MCT knockdown groups, Dunn's multiple comparisons test returned p-values >0.9999 ([Figure 6C](#)), indicating no significant difference from the control. It is worth noting that one of the negative controls, MCT1 RNAi, which was expected to exhibit no phenotype, showed an unexpectedly higher mean number of lysotracker-positive germ cysts (mean=7.61). However, this was primarily driven by two outlier testes, each displaying more than 20 germ cyst deaths, leading to a relatively large SD (SD = 3.89) and a biased mean. Despite this, the p-value of this group remained >0.9999 , confirming no significant difference from the control. Considering the statistics and the fact that MCT1 is not expressed in germ cysts, it was immediately excluded as a potential hit.

To further consolidate our analysis, each experimental group was compared to its corresponding control batch collected on the same day ([Figure 6D](#)). For comparisons involving more than two groups, a Kruskal-Wallis test followed by Dunn's multiple comparison test was conducted, whereas for comparisons between only one experimental group and the relative control, a non-parametric t-test (Mann-Whitney test) was carried out. Specifically, CG10019, Outsiders, and Hermes were analyzed together in one test, while Tarag, MCT1, CG13907, CG8468, CG6006, and Silnoon were grouped in another. *Karmoisin* and CG8389 were analyzed separately using the Mann-Whitney test. While minor variations in the p-value compared with previous tests were observed (CG10019 & *Karmoisin*), the overall significance level for each group remains consistent.

In conclusion, *Hermes* and *Karmoisin* knockdown significantly increases germ cell death as evidenced by the statistically higher number of lysotracker-positive germ cysts. This

preliminary result suggests that Hermes and Karmoisin are the two potential MCT hits that mediate the metabolite shuttling and maintain the survival of the germline.

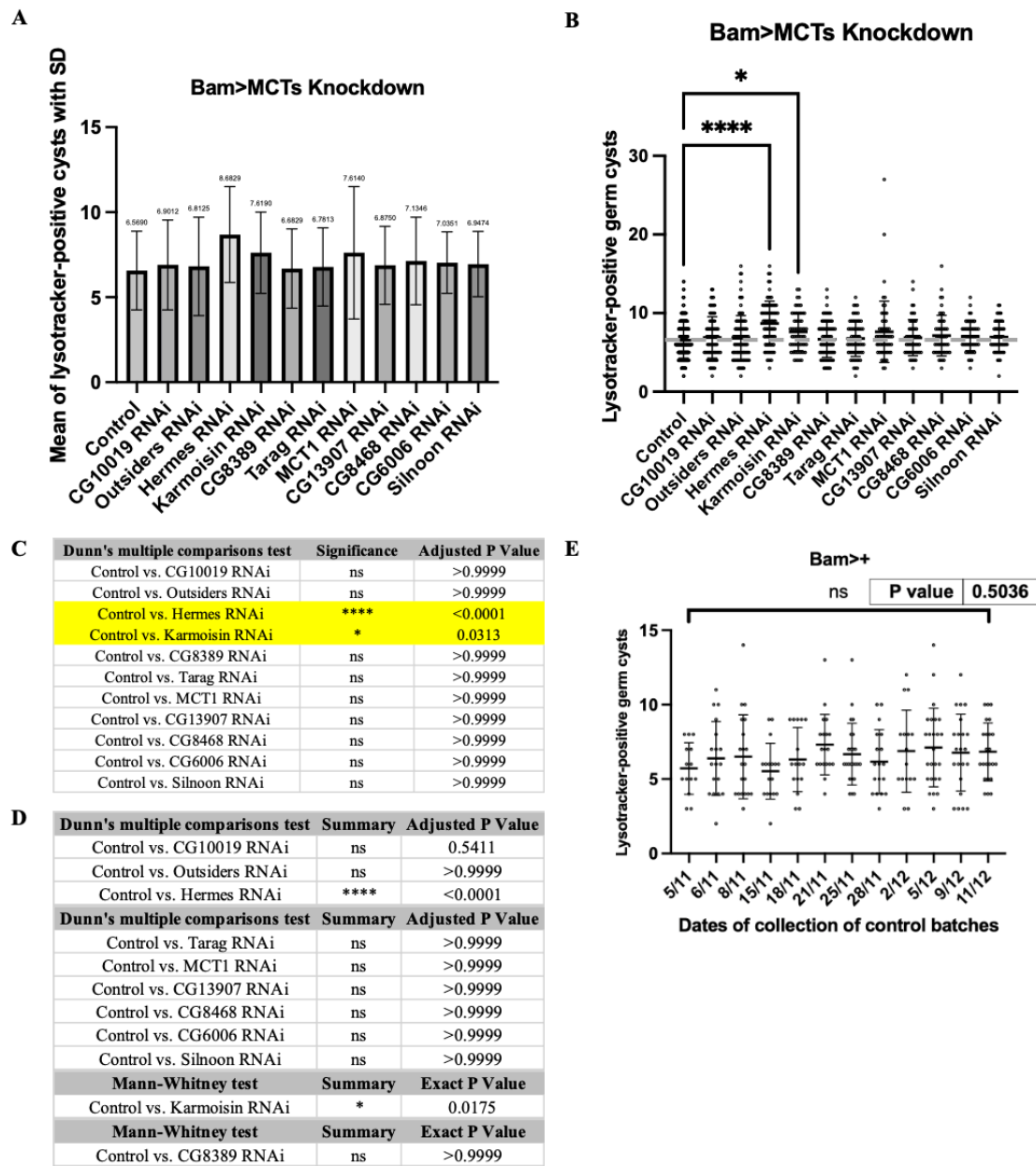


Figure 6 Hermes and Karmoisin Knockdown significantly increased germ cyst deaths. Note – The number of asterisk (*) indicates the level of significance: (1) ns for $p>0.05$, (2) * for $p<0.05$, (3) ** for $p<0.01$, (4) *** for $p<0.001$, (5) **** for $p<0.0001$. Dunn's multiple comparison test is used for pairwise comparisons to determine which specific groups differ after the Kruskal-Wallis test identifies an overall significant difference among them. (A) Bar graph displaying the mean number of the lysotracker-positive germ cysts in MCT knockdown groups and the control with SD as error bars. The mean value for each group is illustrated above the corresponding bar, as well as the level of significance. The pairs that are not significant are neglected. (B) Graph for the Kruskal-Wallis test showing the distribution of the number of lysotracker-positive cysts in each group. Each dot represents an individual testis. The grey dotted line denotes the mean value of the control group. (C) Results of Kruskal-Wallis and Dunn's multiple comparison tests for experimental groups with all control batches. (D) Results of Kruskal-Wallis and Dunn's multiple comparison tests and the Mann-Whitney test for experimental groups in comparison with their corresponding control batches. (E) Graph showing the number of lysotracker-positive germ cysts in 12 control batches with SD as error bars.

4. Discussion and Perspectives

Cell-cell communication in the *Drosophila* testis, especially between the two stem cell lineages (GSCs and CySCs) and their progeny, is well-characterized for its role in coordinating differentiation ([Kiger et al., 2000](#)). However, the metabolic interplay between the germline and somatic lineages remains poorly understood in the *Drosophila* testis. In contrast, metabolic coupling is well-established in the nervous systems of both vertebrates and invertebrates, where glial cells transfer key metabolites to neurons to support neuronal function ([Tsacopoulos et al., 1988](#), [Pellerin and Magistretti, 1994](#)). The astrocyte-neuron lactate shuttle (ANLS) model, first proposed by Pellerin and Magistretti in 1994, demonstrated that glutamate uptake by astrocytes promotes anaerobic glycolysis and lactate production, which serves as a metabolic substrate for neurons during activation ([Pellerin and Magistretti, 1994](#)). This model was further supported by the differential distribution of Lactate Dehydrogenase (LDH) isoenzymes in astrocytes and neurons: LDH1, which favors lactate production, is predominantly expressed in astrocytes, whereas neurons exclusively express LDH5, which preferentially converts lactate to pyruvate for oxidative metabolism ([Bittar et al., 1996](#)). Additionally, MCTs, essential for transmembrane monocarboxylate shuttling, exhibit cell-type-specific expression in rodents - astrocytes express the low-affinity transporters MCT1 and MCT4, while neurons express the high-affinity MCT2, facilitating efficient lactate uptake ([Pierre and Pellerin, 2005](#)). Alongside that, pyruvate to lactate conversion depends on the cytosolic redox state of the cell, with a higher reducing environment favoring lactate production. Studies have shown that astrocytes exhibit a higher NADH to NAD⁺ ratio compared to neurons, indicating that astrocytes are thermodynamically predisposed to lactate formation ([Jakoby et al., 2014](#), [Magistretti and Allaman, 2018](#)). A lactate gradient from astrocytes to neurons was shown

in vivo using a genetically encoded lactate sensor and two-photon microscopy, further reinforcing the metabolic coupling. Trans-acceleration, a unique MCT property, means that the presence of extracellular monocarboxylate stimulates the transporter substrate efflux. When the MCTs in neurons and astrocytes were trans-accelerated using extracellular pyruvate, intracellular lactate levels decreased in the astrocytes but remained unchanged in the neurons, supporting the directionality of lactate transport ([Mächler et al., 2016](#)). Expanding this perspective beyond mammals, studies in invertebrates have shown a similar metabolic interplay. Research by the Schimeier group demonstrated that glial glycolysis is essential for neuronal survival in the *Drosophila* brain, as key metabolites such as alanine and lactate, produced by glia, can be transported via MCTs to fuel the neurons ([Volkenhoff et al., 2015](#)). Additionally, metabolic communication between Sertoli cells and germ cells during mammalian spermatogenesis is well-studied. Mammalian germ cells are enclosed by the tight junctions of Sertoli cells, forming the blood-testis barrier (BTB) ([Crisóstomo et al., 2018](#)). Similarly, in *Drosophila*, germ cells are encapsulated by cyst cells that physically separate them from circulation. Sertoli cells serve as the major lactate source for the germ cells by converting glucose primarily into lactate, which is then exported via MCT1 into the surrounding microenvironment and captured by germ cells via MCT2 ([Rato et al., 2012](#)). Moreover, Sertoli cells can sustain lactate production even under glucose deprivation by utilizing lipid metabolism, as glucose is not their primary energy source. During glucose deprivation, Sertoli cells activate the AMPK pathway, which enhances GLUT1 expression to facilitate glucose uptake and ensures a stable lactate supply for germ cells by upregulating MCT4 (responsible for lactate export) and downregulating MCT1 (involved in lactate recapture in Sertoli cells) ([Galardo et al., 2007](#), [Riera et al., 2009](#)). Similar metabolic interactions and support can also be found in skeletal muscle where white glycolytic fibers generate lactate as a byproduct of anaerobic glycolysis, which is

subsequently utilized by red oxidative fibers to fuel aerobic respiration ([Brooks, 2002](#)). Collectively, these findings underscore the evolutionary conservation of metabolic coupling, wherein specialized support cells provide essential metabolites to sustain the function and viability of adjacent cell populations across diverse tissues and organisms via metabolite transporters. These findings also highlighted lactate as an important bridge between anaerobic and aerobic metabolism. This aligns with our previous discovery in the lab, where knocking down components of the glycolytic pathway in cyst cells revealed that lactate is the primary metabolite transported from the somatic cysts to the germline in the *Drosophila testis*.

Our two potential hits, Hermes (CG11665) and Karmoisin (CG12286), are both members of the SLC16 family of monocarboxylate transporters, predicted to have 12 transmembrane domains (TMDs) with intracellular N and C terminus and a large intercellular loop between TMD 6 and 7 ([Halestrap and Meredith, 2004](#)).

The MCTs are categorized into two groups: proton-linked MCTs (MCTs 1-4) and hydrophobic MCTs. The proton-linked MCTs possess key residues (K38, D309, and R313) crucial for monocarboxylate transport ([Halestrap, 2013](#), [Wilson et al., 2009](#)). Sequence alignment of *CG11665* with human proton-linked MCTs shows conservation of critical motifs, confirming its role in proton-coupled metabolite shuffling. *CG11665* was first named *Hermes (Hrm)* in 2018 after its important function in autophagy during cell death was identified. The same study also validated that Hermes (Hrm) is a proton-coupled MCT primarily involved in pyruvate transport rather than lactate transport. By using a pH-sensitive fluorophore to label the glycocalyxes of HEK293T cells, the researchers were able to measure proton transport, which reflects the rate of metabolite transport. Both *rMCT1* and *Hermes* expression enhanced proton-coupled pyruvate transport, with a faster influx rate when extracellular pyruvate was perfused and a faster efflux rate when pyruvate was

washed out, compared to control cells. However, unlike rMCT1, the *Hermes* expression had no significant effect on lactate transport ([Velentzas et al., 2018](#)). This study suggests that Hermes preferentially transports pyruvate over lactate, which seems contradictory to our preliminary result that Hermes is involved in the lactate shuttling from somatic cells to the germ cysts. The conclusion that lactate is the key metabolite transported, is supported by the following findings in the lab: (1) Ldh knockdown in the soma increases germ cell death, (2) Ldh knockdown in the soma increases pyruvate levels in the soma (validated using the pyruvate reporter, pyrSF), and (3) Ldh knockdown in the soma decreases lactate levels in both the soma and germline (validated using the lactate reporter, CanlonicSF). The difference between our findings and those of Velentzas et al. may stem from differences in experimental focus rather than a direct contradiction. Our study specifically assumes that germ cell survival is only linked to lactate metabolism. The distinction may be attributed to two factors: (1) Hermes RNAi might have off-target effects, and (2) the role of Hermes in the *Drosophila* testis may differ from its function observed in the human HEK293T cell line, suggesting species- or tissue-specific differences in its metabolic role. For future investigations, we plan to validate that the observed increase in germ cell death is not due to an off-target effect by using a second Hermes RNAi line. If the results from the second RNAi line are consistent with the initial finding, then a Hermes knockout mutant clone should be generated to further confirm the validity of the hit. Additionally, lactate reporters will be expressed in the germline to measure lactate levels, helping to determine whether lactate import is inhibited upon Hermes knockdown.

Additionally, SLC16A11, the most similar human ortholog of Hermes, alters lipid metabolism in hepatocytes when diminished and has been associated with type 2 diabetes ([Rusu et al., 2017](#)). Similarly, the deletion of Hermes in the *Drosophila* salivary gland leads to alterations of the lipid metabolism, characterized by having enlarged lipid droplets

([Velentzas et al., 2018](#)). However, it remains unclear whether Hermes knockdown has a similar effect on lipid metabolism in germ cells, and its potential impact on germ cell survival has yet to be investigated.

The second hit, *Karmoisin* (*CG12286*), was not named based on its function in metabolite transport but rather due to its role in ommochrome biogenesis, which is the brown pigment in the *Drosophila* eye. *Karmoisin* encodes for Phenoxazinone Synthetase (PHS), the terminal step enzyme for ommochrome formation, which catalyzes the condensation of 3-hydroxykynurenine to xanthommatin ([Liu et al., 2019](#), [Grubbs et al., 2015](#)). *Karmoisin* is orthologous to human SLC16A10 (MCT10) and partly to SLC16A2 (MCT8) based on sequence and structural similarity. MCT10, also known as T-type amino acid transporter 1 (TAT1), is a sodium and proton-independent aromatic amino acid (tryptophan, tyrosine, phenylalanine) transporter ([Kim et al., 2002](#)). In TAT1 null mice (*tar^{-/-}*), the levels of all three aromatic amino acids are higher than in the control ([Mariotta et al., 2012](#)), indicating failure of the import of these amino acids. This was also validated in yeast, where the N81K mutation in TMD1 of MCT10 abolished its tryptophan import activity ([Uemura et al., 2017](#)). What's more, MCT10 shares homology with MCT8, which is a highly specialized thyroid hormones transporter, although MCT10 only exhibits reduced thyroid hormone transport activity ([Friesema et al., 2003](#)). Therefore, based on these structural and sequence similarities, the metabolites transported by *Karmoisin* are predicted to be similar to MCT10, primarily transporting aromatic amino acids. Limited evidence has shown that *Karmoisin* is involved in transmembrane lactate transport. Therefore, a second *Karmoisin* RNAi line, *Kar* knockout mutant clone, lactate reporter should be used to test the robustness of our results. Due to time constraints, I was unable to complete the follow-up experiments investigating both Hermes and *Karmoisin*.

After lactate is transported into germ cells, the next key question in our hypothesis is how germ cells metabolize the lactate provided by somatic cysts to support their survival. Evidence from human spermatogenesis suggests that lactate is converted into pyruvate by testis-specific lactate dehydrogenase C (LDHC) and then participates in other downstream metabolic and signaling pathways ([Boussouar and Benahmed, 2004](#), [Goldberg, 1989](#)). For example, Lactate has been shown to exert an anti-apoptotic effect on germ cells, primarily in spermatocytes and spermatids, with a lesser impact on spermatogonia. Dysregulation of lactate levels has been associated with male infertility ([Erkkilä et al., 2002](#)). However, whether the germ cell deaths in *Drosophila* testis are related to the anti-apoptotic effect of lactate remains unclear, but a similar lactate-pyruvate conversion appears to be present in the *Drosophila* testis. Our lab performed an *Ldh* knockdown using the *Bam-Gal4* driver in late-stage germ cells, which resulted in a significant increase in germ cell death. Additionally, fluorescence imaging of a lactate reporter revealed an accumulation of lactate within the germ cells, indicating that impaired lactate-pyruvate conversion affects germ cell survival.

To further elucidate the metabolic fate of pyruvate in germ cells, four potential pathways should be considered: (1) conversion to acetyl-CoA by *Pdha* for entry into the TCA cycle ([Patel et al., 2014](#)), (2) conversion to oxaloacetate by Pyruvate carboxylase (*Pcb*) ([Jitrapakdee and Wallace, 1999](#)), (3) transamination to amino acids such as alanine ([Felig, 1973](#)), (4) conversion to malate by malic enzyme (*Men*) ([Wise and Ball, 1964](#)). *Pdha* knockdown with *Bam-Gal4* was conducted and did not increase germ cell death, suggesting that pyruvate is not converted to acetyl-CoA and the germline does rely on TCA. This was supported by François et al., who demonstrated that knocking down 32 genes encoding enzymes involved in all eight steps of the TCA cycle using *Nanos-Gal4* driver had no impact on male fertility. Furthermore, CRISPR-Cas9 was used to generate loss-of-function

mutations in two genes encoding citrate synthase (*dCS* and *CG14740*), the enzyme that catalyzes the first committed step of the TCA cycle by converting oxaloacetate and acetyl-CoA into citrate. To rule out functional redundancy, a double knockdown was also performed. Neither single knockout nor double knockdown affected male fertility, confirming that TCA is dispensable for germline survival ([François et al., 2023](#)). However, as the study by François et al. (2023) utilized the early germ cell driver nanos-Gal4, further validation using Bam-Gal4 to knock down TCA cycle enzymes in late-stage germ cells is necessary. Additionally, knockdown of transaminases, including Alanine transaminase (Alat) and Serine-pyruvate transaminase (Spat), which convert pyruvate to alanine, as well as Pyruvate carboxylase (Pcb), which converts pyruvate to oxaloacetate, did not affect germ cell survival. These results suggest that pyruvate is not primarily metabolized into alanine or oxaloacetate in germ cells (Sainz de la Maza, unpublished results). Malic enzyme (Men) catalyzes the decarboxylation of malate to pyruvate, producing NAD(P)H from NAD(P)⁺ ([Wise and Ball, 1964](#)). This reaction is reversible, with malate synthesis from pyruvate occurring when the reverse reaction is favored. The reverse reaction is more likely to occur under conditions of high pyruvate and a low NAD(P)⁺/NAD(P)H ratio ([Frenkel, 1975](#)). In germ cells, where the TCA cycle is dispensable, the NADH/NAD⁺ ratio is expected to be low. This low NADH level would promote the reverse reaction of malic enzyme, favoring the conversion of pyruvate to malate. Therefore, if pyruvate in the germline is converted to malate, the *Men* knockout is expected to increase germ cell death. However, surprisingly, *Men* knockdown did not result in a phenotype, while knockdowns of *Menl1* and *Menl2* (malic enzyme-like 1/2) did produce a phenotype. This may be due to two potential reasons: either the *Men* RNAi was not effective or *Menl1/2*, instead of *Men*, is the primary enzyme responsible for this conversion in the germline. This preliminary result showed that

pyruvate is converted to malate in the germline and how the germline used malate to support their survival requires further investigation.

This project filled the gap in lactate shuttling from the somatic cysts to the germ cysts by discovering the monocarboxylate transporter responsible for lactate import in the germ cells (Hermes and Karmosin). However, these findings require further validation using a second RNAi line and mutant clones, together with expressing lactate reporters in germ cells. Future work should focus on further elucidating the germline metabolism, as investigating this in the *Drosophila* testis may offer valuable insights into how metabolism influences cell fate and behavior, as well as how inter-lineage metabolic communication is critical in tissue homeostasis and development.

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Appendices

Appendix 1

MCT	Function	Family
CG10019	Predicted to enable transmembrane transporter activity. Predicted to be involved in transmembrane transport. Predicted to be active in plasma membrane.	SLC16
Hermes	Enables pyruvate secondary active transmembrane transporter activity. Involved in plasma membrane pyruvate transport. Located in plasma membrane.	SLC16
Karmoisin	Predicted to enable L-amino acid transmembrane transporter activity. Predicted to be involved in aromatic amino acid transport. Predicted to be active in plasma membrane.	SLC16
Tarag	Enables acetate transmembrane transporter activity. Involved in acetate transmembrane transport. Predicted to be active in plasma membrane.	SLC16
Outsiders	Predicted to enable monocarboxylic acid transmembrane transporter activity, possibly lactate. Involved in germ cell development. Predicted to be active in plasma membrane.	SLC16
CG8389	Predicted to enable monocarboxylic acid transmembrane transporter activity. Predicted to be involved in monocarboxylic acid transport. Predicted to be active in plasma membrane.	SLC16
CG8034	Predicted to enable monocarboxylic acid transmembrane transporter activity. Predicted to be involved in monocarboxylic acid transport. Predicted to be active in plasma membrane.	SLC16
Prestin	Plasma membrane transmembrane anion transporter that mediates the electrogenic exchange of chloride/bicarbonate, chloride/oxalate, chloride/sulfate and chloride/formate.	SLC26
CG8051	Predicted to enable monocarboxylic acid transmembrane transporter activity. Predicted to be involved in monocarboxylic acid transport. Predicted to be active in plasma membrane.	SLC16

Appendix 1 MCTs and their predicted functions (Fly Cell Atlas)